Raphaël Magron-Lorenzo

Luxembourg | raphael.mglo23@gmail.com | +352 621 262 971 | rm91.github.io | linkedin.com/in/rml9/

EDUCATION

University of St Andrews, United Kingdom (2023–2027)
MPhys (Hons) Mathematics and Theoretical Physics

- On track for First Class Honours

Sportlycée, Luxembourg (2020–2023)
Section B: Mathematics and Computer Science

- Baccalaureate: 54/60 (Highest Distinction, Rank in school: 1st)

RESEARCH EXPERIENCES

University of St Andrews - Analysis Department (2025)
Supervisor: Dr Natalia Jurga. Funding: The Carnegie Trust & StARIS Scheme (total: £2900)

- Formalised classical electrodynamics using the framework of dynamical systems and random walks on networks
- Extended the Snell-Doyle framework from discrete Markov chains to irreversible networks
- Analysed the metricity of electrical networks via the 'effective resistance'
- Studied the convergence of sequences of electrical networks & investigate the metricity of the limit space

University of St Andrews - Astrophysics Department (2025)
Supervisor: Prof. Keith Horne. Funding: Unfunded.

- Investigated conformal gravity models with a focus on Nesbet's formulation
- Incorporated variations of the Ricci scalar into the field equations
- Performed numerical analysis and visualisation of cosmological implications

University of Luxembourg - *Educational Project in Physics* (2022)
Supervisor: Prof. Ludger Wirtz

- Developed educational resources on harmonic oscillators for advanced high-school students
- Introduced computational physics tools to illustrate analytical and numerical approaches

RESEARCH INTERESTS

- **Pure mathematics -** problems motivated by physical intuitions: potential theory, thermodynamic formalism, spectral theory, stochastic processes, symplectic geometry
- **Computational Physics -** Numerical methods and modelling for analytically intractable systems, with a focus on numerical integration and combining analytical insight with efficient computational approaches

DISTINCTIONS & PRIZES

- Class Medal - Best Student in Theoretical Physics, University of St Andrews (2024-2025)
- William David Brodie Prize, University of St Andrews (2024-2025)
- Dean's List, University of St Andrews (2023-2025)
- Belgian Mathematics Olympiad - Semi-finalist (2022)
- Student Merit, Sportlycée Luxembourg (2017-2023)

SKILLS & LANGUAGES

- **Technical –** Python (OOP, SciPy, NumPy, Matplotlib), LaTeX, Git
- **Scientific communication** - Reports, Presentations, Code Documentation
- **Microsoft Office** - Good proficiency: Word, Excel, PowerPoint
- **Languages -** French, Luxembourgish (native); German, English (fluent); Spanish, Japanese (JLPTN4, beginner)

EXTRA-CURRICULAR

- **Previous jobs** - Postman (POST Luxembourg: 2023, 2024)
- **Sport** - Squash (Uni of St Andrews 1st team Captain, Luxembourgish National Squad, World Championships)
- **Music** - Music Theory (all grades), Recorder (all grades), and Piano (beginner)
- **University societies** - Film, Debate, Philosophy, Chess